Machine Learning Analysis of Marketing and Product Performance

Dataset

- Name: Marketing and Product Performance Dataset
- Source: Kaggle
- Rows: ~1000+
- Columns: ~20+

Dataset Link

https://www.kaggle.com/datasets/imranalishahh/marketing-and-product-performance-dataset

Target Variable

High_Performance (derived from Revenue_Generated)

Task Type

Classification

Problem Statement

To predict whether a product or campaign will achieve high performance based on marketing and product-related features.

Research Questions

1. **How do baseline models perform?**
2. **Which model performs best?**
3. **Does preprocessing improve performance?**
4. **Which features are important?**
5. **How do evaluation metrics compare?**
6. **Is the model robust?**
7. **Which model should be selected?**

Methodology

- **Data cleaning and preprocessing**
- **Feature encoding**
- **Train-test split**
- **Model training and evaluation**

Models Used

- **Logistic Regression**
- **Decision Tree**
- **KNN**
- **Random Forest**
- **XGBoost**
- **SVM**

Evaluation Metrics

- **Accuracy**
- **Precision**
- **Recall**
- **F1-score**

Expected Outputs

- **Tables (CSV)**
- **Graphs (PDF)**
- **Feature importance plots**